# PIXELROOT: Hardware-Anchored, Sensor-Level Media Provenance with Real-Time Blockchain Notarization for Trustworthy Imaging

Mihir Kavishwar

***Abstract*—The realism of AI-generated "deepfake" imagery has outpaced society's ability to tell synthetic media from camera-captured reality. Detectors generalize poorly to unseen generators, and provenance schemes that bind trust to *metadata* fail once the metadata is stripped. We present PIXELROOT, which moves the root of trust into the *pixels* and onto a public blockchain. At capture, a CMOS sensor embeds a $128$-bit, Reed–Solomon-coded provenance payload through a keyed, low-frequency per-region gain modulation. Each sensor holds a signing key generated on-die and certified on the blockchain by its manufacturer; it signs a capture record that binds the payload to the frame's hashes and to a robust content fingerprint, and the record is notarized on the same blockchain, which also records time-scoped revocation. We contribute a formal architecture and threat model with five security propositions; a measured robustness study on the Kodak images with real codecs, recovering the payload on $22$–$23/24$ images down to JPEG $Q20$ ($21/24$ at $Q10$) at $38$ dB PSNR with $0/1224$ false accepts; a device-identity evaluation showing that the in-pixel mark alone can be copied without any key ($78\%$ of trials) while the content-bound record rejects every cross-scene copy and large splice on held-out images ($0/180$) and lets $90\%$ of processed genuine copies verify, linking local edits back to the original; a tested registry contract ($47$k gas per batch, $O(1)$ per image); a CMOS-manufacturing feasibility analysis of per-pixel gain; and a privacy-preserving notarization design with a phased adoption path toward a cross-OEM interoperability standard.**

***Index Terms*—media provenance, deepfakes, blockchain, CMOS image sensor, PRNU, digital watermarking, content authenticity, C2PA, image forensics, JPEG.**

## I. INTRODUCTION

DIGITAL images and videos have become primary evidence in journalism, courts, insurance, humanitarian monitoring, and everyday social discourse. That evidentiary role is collapsing. Diffusion models, generative adversarial networks, and neural rendering now synthesize photorealistic faces, scenes, and full videos that humans—and automated detectors—routinely fail to distinguish from reality [1], [2]. The harm is no longer hypothetical. Industry incident trackers attribute on the order of $1.2–1.3 billion in documented fraud losses to deepfakes in 2025, rising steeply year over year, with the overwhelming majority of losses originating on social-media platforms [3], [4]; enterprise surveys report that a majority of organizations experienced deepfake-related incidents with mean losses exceeding $280,000 [5]. Beyond fraud, non-consensual intimate imagery and election disinformation erode personal safety and institutional trust.

The dominant technical response has been *reactive detection*: train a classifier to spot the statistical fingerprints of a given generator. This strategy is structurally disadvantaged. Detectors overfit to the artifacts of the generators in their training set and degrade sharply on unseen models—recent in-the-wild benchmarks report AUC drops of roughly $45$–$50\%$ relative to academic datasets [1], [6]. Because every published detector also serves as a differentiable signal that the next generation of generators can be trained to defeat, detection is a perpetual "tug of war" that the defender is not guaranteed to win [7], [8].

An alternative philosophy inverts the problem: instead of trying to prove a piece of media is *fake*, prove that a piece of media is *real* by binding it, at creation time, to a trustworthy origin. This is the *provenance* paradigm now embodied by the C2PA Content Credentials standard and by secure-capture products such as Truepic and ProofMode [9]–[12]. These systems are a major advance—C2PA supports hardware-backed signing, rich edit-history assertions, and ingredient hashes, and our goal is to *complement* rather than replace it. But the authenticity claim is carried, in the common case, in a *signed metadata container* attached to the file, and that container is a single point of failure: metadata can be deliberately stripped by platforms, lost in transcoding, or, worse, the binding itself can be attacked. In September 2025, a vulnerability in the Nikon Z6 III's C2PA implementation let a camera sign fabricated composites as authentic, forcing the vendor to suspend its authenticity service and revoke every certificate it had issued [13]. If the credential is removed, the pixels carry no inherent evidence of their origin. PIXELROOT targets exactly this gap, and can populate C2PA assertions while adding an in-pixel, ledger-witnessed binding that survives container loss.

### A. The PIXELROOT thesis

We argue that a robust root of trust for visual media should live where the information actually is: in the pixels, anchored to physical hardware, and witnessed by an immutable public ledger. PIXELROOT embeds a provenance payload *into the pixel values* at the moment of capture through a keyed, low-frequency *per-region* sensor gain modulation—a perturbation small enough to be imperceptible yet structured to survive lossy re-encoding—and *simultaneously* notarizes a

M. Kavishwar is an independent researcher with the PixelRoot Project (e-mail: hello@mihirkavishwar.com).

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cryptographic commitment of the image to a blockchain. This yields two independent bindings:

- a **hard binding**—a SHA-256 commitment recorded on-chain, giving an immutable, publicly verifiable timestamp of existence; and
- a **soft, in-pixel binding**—a sparse, error-corrected signature tied to the sensor's factory identity that travels *with the pixels* even if the file container and its metadata are destroyed.

Both bindings are anchored to the **sensor's own identity**: a signing key generated inside the sensor at manufacture, never exported (not even to the manufacturer), and certified on the same blockchain. The sensor signs a capture record that ties the in-pixel payload to the frame's hashes and to a robust fingerprint of its content, so a payload copied onto other content no longer verifies (§IV-G).

The two bindings are complementary: the hard binding gives cryptographic non-repudiation but is destroyed by any re-encoding; the in-pixel binding tolerates mild processing and survives metadata stripping, but is statistically softer. Together they cover each other's failure modes.

### B. Contributions

1) **An architecture** (§IV) with a feasibility-grounded, tiered hardware ask: a keyed, error-corrected payload written by per-region sensor gain, and the CMOS routes that make per-pixel gain manufacturable.
2) **A device-identity layer on the ledger** (§IV-G, §IV-H): on-die sensor keys certified by lot, time-scoped revocation, sensor-signed records bound to a content fingerprint, a block anchor that bounds capture time from below, and a tested registry contract.
3) **A security analysis** (§III, §V) with five propositions and a threat catalogue, including the watermark copy attack, sensor-key theft, and adaptive local edits.
4) **Measurements on real images** (§VII): robustness of the in-pixel channel to real codecs and edits, and an end-to-end identity evaluation on held-out images showing that the mark alone is copyable without a key and that the content-bound record stops cross-scene forgery.
5) **A deployable design** (§VIII): verification from public data, privacy-preserving notarization, and a phased adoption path.

## II. Related Work and Literature Survey

We organize prior art into five threads and then identify the gap PixelRoot fills.

Four lines of work bear on PixelRoot. *Passive sensor forensics* attributes an image to a camera through its photo-response non-uniformity (PRNU) fingerprint [14], [15], but needs reference images, weakens under compression [16], and can be copied onto forged images [17]. *Watermarking* embeds recoverable signals [18]–[20] and now marks generated content at scale [21]–[23], yet marks can be copied [24], averaged out [25] or removed by regeneration [26], [27]; in-camera marking has precedent [28]. *Provenance standards* such as C2PA bind signed manifests to assets and now ship in cameras and phones [9]–[12], but manifests are stripped in distribution and revocation has failed in practice [13]. *Blockchain media integrity* anchors hashes on public ledgers [29]–[31], typically after capture and in software. The extended version surveys each line in detail.

TABLE I
Where prior art sits, and the gap PixelRoot targets.

| Approach | HW root | Surv. meta-strip | Public ledger | Affirm. proof | Real-time |
|---|---|---|---|---|---|
| Deepfake detectors [1] | ✗ | — | ✗ | ✗ | ✓ |
| PRNU forensics [14] | part. | ✓ | ✗ | part. | ✗ |
| Watermarking [18] | ✗ | ✓ | ✗ | part. | ✓ |
| C2PA / Truepic [9], [10] | ✓ | part.† | opt. | ✓ | ✓ |
| Blockchain hashing [29] | ✗ | ✗ | ✓ | ✓ | part. |
| **PixelRoot (this work)** | ✓ | ✓ | ✓ | ✓ | ✓ |

HW root = hardware/sensor root of trust; Surv. meta-strip = survives metadata removal; Affirm. proof = affirmative proof of authenticity.
†C2PA 2.x can recover a stripped manifest through a registered watermark or fingerprint (soft binding) when one is used.

### A. Gap analysis and precise novelty

Table I summarizes the landscape. Each ingredient exists in isolation: watermarking embeds recoverable signals in pixels; PRNU ties images to a physical sensor; C2PA provides hardware-backed signed provenance; and prior systems anchor media hashes to a blockchain [29]; even in-camera watermarking has precedent [28]. Our claim is therefore *not* that any single component is new. The specific, defensible novelty of PixelRoot is their *co-design*: a **(i) hardware-rooted, key-protected, error-corrected signal embedded by the sensor's own gain path**—so the binding survives metadata stripping and is rooted in silicon rather than a software key—that is **(ii) bound at the shutter to a public-ledger commitment**, yielding two independent failure-disjoint bindings, **(iii) realized through a *feasibility-grounded* per-column/per-region gain primitive** we ask manufacturers to standardize rather than the per-pixel control prior "sensor-embedding" sketches assumed, and **(iv) bound to a sensor key certified and revocable on the same public ledger**, through a signed, content-bound record that makes the in-pixel identity non-transferable. No prior approach occupies the intersection of hardware root, metadata-strip survival, and public real-time notarization. We claim contributions at the level of architecture, the concrete hardware ask, the measured robustness of the carrier, and the privacy design—not the invention of watermarking or blockchains.

## III. Threat Model and Design Goals

### A. Assumptions

We assume: (A1) the sensor's OTP identity and gain registers are written at manufacture and are not feasibly reprogrammable in the field without specialized equipment; (A2) the capture-time signing/notarization environment is attested (e.g. via a secure element), as in current hardware-backed

C2PA deployments [10], [12]; (A3) the blockchain is append-only and its history is economically infeasible to rewrite within the relevant timeframe [32]; (A4) standard cryptographic primitives are secure [33], [34]; (A5) each sensor generates its signing key on-die at manufacture and the private key cannot be exported without an invasive attack on that physical device; manufacturers admitted to the registry certify only keys produced on their lines.

### B. Adversary

The adversary may: forge or edit file metadata; generate a deepfake or Photoshop a real image; re-encode/transcode/compress media; re-photograph a screen (analog "re-capture"); attempt to estimate and transplant a victim's sensor fingerprint (PRNU-copy [17]); and replay or back-date claims; read any published capture record; estimate and copy an in-pixel mark from images it obtains (with or without the original); and extract the key of a device it physically possesses. The adversary cannot economically clone a specific physical CMOS die, extract keys from devices it does not hold, or rewrite ledger history.

### C. Design goals

**G1 Affirmative authenticity:** produce verifiable evidence that media originated from a genuine sensor at a specific time/place. **G2 Hardware root:** bind the proof to physical sensor identity, not to software-only secrets. **G3 Metadata-independence:** retain evidentiary value even if the file container/metadata is stripped. **G4 Public verifiability:** any third party (platform, court, viewer) can verify a capture from public data—the ledger, the published record, and the OEM's carrier key—without contacting the originator or possessing the camera (§VIII). **G5 Minimal capture latency:** submit the commitment at capture, so the sealing delay is the batch interval plus the chain's block time (seconds on low-latency chains and rollups); we measure gas but leave public-network latency to future work (F4). **G6 Imperceptibility:** embedding must not visibly degrade the image. **G7 Standardizability:** be deployable as a cross-OEM interoperability standard.

## IV. The PixelRoot Architecture

PixelRoot comprises a capture-time *embed-and-notarize* pipeline and a *verification* pipeline. Fig. 1 shows the end-to-end architecture: the upper band is the on-device capture path that produces the two bindings, and the lower band is the third-party verification path that consumes them.

### A. Hardware feasibility: the gain-control spectrum

A central objection to any "sensor-level embedding" proposal is that commercial CMOS sensors do *not* expose arbitrary per-pixel gain. PixelRoot does not require it. We organize sensor gain controllability into a spectrum and target the realistic middle of it: from a single global gain (too coarse to carry a payload), through per-column programmable-gain amplifiers (present in column-parallel readout) and per-region conversion-gain or HDR zones [35]–[37], to per-pixel gain (manufacturable but not yet productized, §IV-B). A static per-column profile gives every block in a column the same chip sign, so the block-keyed layout needs per-region control or per-column gains re-programmed at line rate.

**The PixelRoot ask (tiered).** We ask manufacturers for a *provenance-write* capability, achievable at three increasing cost points: (1) a programmable *per-column* gain vector and a coarse *per-region* tile map addressable at capture—realizable on conventional sensors today; and, as the apex, (2) *per-pixel* gain via the manufacturing routes of §IV-B. All tiers reuse the OTP identity and attested clock that hardware-backed C2PA pipelines already ship [10], [12]. PixelRoot's robust carrier needs only a *low-frequency gain ramp over an image region* (§IV-E), which per-region control realizes, as does per-column control re-programmed at line rate; per-pixel control is strictly more general—it *subsumes* per-column and per-region patterns and additionally enables a fine-grained tamper-localization layer (§IV-B).

**Software-ISP fallback (legacy).** Devices without any gain capability can apply the identical modulation in the attested image-signal processor (ISP) immediately after readout. This preserves the in-pixel binding and notarization but *weakens the hardware root*: the modulation is imposed in firmware rather than the photo-electron domain, so it is only as trustworthy as the attested execution environment (comparable to today's secure-capture apps). We label such captures at a lower assurance tier (§VIII) and never conflate them with sensor-rooted captures.

### B. Manufacturing per-pixel gain

Per-pixel gain is not exposed by today's commodity sensors, but it is an integration choice rather than a physics barrier: three process technologies in volume production make it achievable—per-pixel conversion-gain selection in dual/triple-conversion-gain pixels [35], [37], coded-exposure pixels with in-pixel memory [38], [39], and 3D-stacked digital pixel sensors joined by Cu–Cu hybrid bonding [40], [41]. The stacked logic tier is also a natural home for the on-die key of §IV-G. Per-pixel control further enables a high-resolution integrity layer that localizes splices (block ROC-AUC $0.97$ on $48$ real splices). The feasibility analysis with engineering schematics and costs, and the localization experiment, are in the extended version supplied as supplementary material.

### C. Provenance payload

The payload $\mathbf{m}$ is a fixed-width record (PixelRoot v2: $128$ bits):

$$\mathbf{m} = \underbrace{\mathrm{ID}}_{48} \,\|\, \underbrace{t}_{32} \,\|\, \underbrace{(\phi, \lambda)}_{48}, \tag{1}$$

where ID is, *by default*, a per-device *pseudonymous* attestation identifier issued by the secure element (a rotating $48$-bit token, not a global serial number, and unlinkable to the physical device without OEM cooperation), $t$ is a Unix timestamp from an attested RTC/NTP source, and $(\phi, \lambda)$ is an *opt-in*, off-by-default quantized latitude/longitude ($24$ bits per coordinate, $\approx 2 \times 10^{-5}$ degree resolution). To protect the journalists and

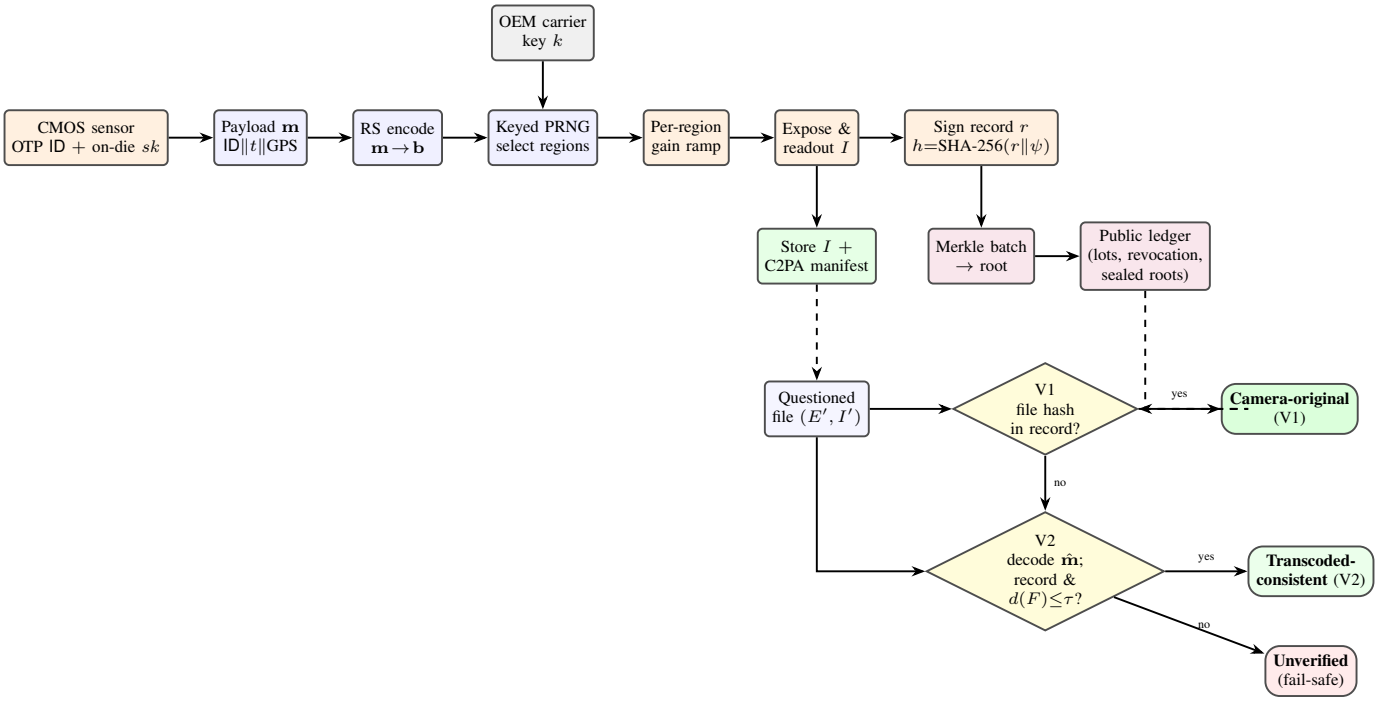



Fig. 1. PIXELROOT architecture. *Top:* the capture path emits a hard binding (the hash of a sensor-signed capture record, sealed on the public ledger) and a soft in-pixel binding (the keyed, error-corrected gain pattern carried inside $I$); the same ledger certifies and revokes sensor keys. *Bottom:* verification checks V1 (the file's hash is committed in a valid record) and V2 (the decoded payload resolves to a valid record whose content fingerprint matches) and returns the strongest supported level, failing safe to "unverified".

activists PIXELROOT aims to serve, the recommended default is pseudonymous ID with *no* location; precise location, when enabled, may be coarsened or replaced by a salted commitment disclosed only under subpoena (§VIII).

### D. Keyed region selection and spread-spectrum coding

PIXELROOT partitions the frame into $8\times 8$ blocks aligned with the JPEG grid and treats *blocks*, not single pixels, as carriers. The carrier layout is derived from the OEM *carrier key* $k$,

$$s = \text{SHA-256}(k), \tag{2}$$

which keys a cryptographic PRNG (a stream/CTR-mode keyed generator) that draws a permutation of the $N_B$ blocks of the frame, so the layout is fixed for a given key and frame size. The layout deliberately does *not* depend on the payload $\mathbf{m}$: a verifier that holds only the pixels must be able to locate the carriers and *read* $\mathbf{m}$, which is what lets the in-pixel binding survive metadata stripping. (The released prototype seeds NumPy's legacy generator with a 32-bit reduction of $s$; a deployment uses the full-width keyed PRNG.) For each coded bit the PRNG draws $R$ distinct carrier blocks without repetition and assigns each an antipodal *chip* $c_j \in \{\pm 1\}$ (spread spectrum). The 128 payload bits are first protected by error correction (a Reed–Solomon code over $\text{GF}(2^8)$ [42]); the repetition factor $R$ gives each bit redundancy across many spatially dispersed blocks. A party without $k$ that *reads* a marked image obtains a random bit string (Prop. 2). The layout is, however, the same for every frame of a given size, and the detector's statistic is computable from public information, so an attacker *can* estimate the mark from marked images and copy or subtract it (§VII-C). PIXELROOT therefore does not rely on the secrecy of $k$ for authenticity: the in-pixel payload is a *pointer* whose meaning is fixed by a sensor-signed, content-bound record on the ledger (§IV-G). Copying the pointer onto other content yields a verdict of UNVERIFIED, and removing it yields the same fail-safe result.

### E. Region-gain (spread-spectrum) embedding

The single-pixel, $2\%$ gain perturbation of naïve designs is a poor carrier: an isolated spike is high-frequency energy, precisely what JPEG discards, and it cannot be read back reliably against textured scene content. PIXELROOT instead embeds each bit as a *low-frequency gain ramp over its carrier blocks*. Let $\Phi_{u,v}(x,y) = \cos\frac{(2x+1)u\pi}{16}\cos\frac{(2y+1)v\pi}{16}$ be a low-order $8\times 8$ DCT basis (we use a low horizontal frequency, $(u,v)=(0,2)$, whose shape a per-column gain profile produces). For a carrier block $B_j$ with chip $c_j$ carrying bit $b\in\{0,1\}$ with sign $\sigma = 2b-1$, the sensor adds

$$\Delta_j(x,y) \;=\; A\,\sigma\,c_j\,\Phi_{u,v}(x,y), \tag{3}$$

i.e. a smooth gain ramp of peak amplitude $A$ gray levels (about $3\%$ of mid-gray at $A=4$). Equivalently, the chosen low-frequency DCT coefficient of $B_j$ is shifted by $\pm A$. This reconciles two domains: the modulation is applied as a *spatial* per-region gain (hardware-realizable, §IV-A) but lives in a *low-frequency DCT* band (JPEG-survivable). Because $\Phi_{u,v}$ for $(u,v)\neq(0,0)$ is zero-mean over the block, the embedding leaves block brightness essentially unchanged and

is imperceptible (measured mean PSNR $38.2\,\mathrm{dB}$ at the $A=4$ operating point and $41.2\,\mathrm{dB}$ at $A=3$, §VII).

*a) Correlation read-back:* A questioned image $I'$ is decoded by *projecting* each carrier block onto the same basis—no per-pixel thresholding and, crucially, no estimate of the block's "original" brightness is needed, because the AC basis is orthogonal to the (content-dominated) DC term. For block $B_j$,

$$\rho_j \;=\; \frac{\langle I'_{B_j},\, \Phi_{u,v}\rangle}{\langle \Phi_{u,v},\, \Phi_{u,v}\rangle} \;\approx\; A\,\sigma\,c_j \;+\; \eta_j, \tag{4}$$

where $\eta_j$ is the host-content and noise term. Because neighbouring blocks of natural images have similar $(0,2)$ coefficients while the chips are independent and zero-mean, the detector first high-passes the field of projections, $\tilde{\rho}_j = \rho_j - \overline{\rho}_{\mathcal{N}(j)}$, where $\overline{\rho}_{\mathcal{N}(j)}$ is the mean over the $5\times 5$ blocks around $B_j$; this removes most of the slowly varying host term. De-spreading and combining the $R$ blocks of a bit,

$$D \;=\; \sum_{j=1}^{R} c_j\,\tilde{\rho}_j \;\approx\; R\,A\,\sigma \;+\; \textstyle\sum_j c_j\tilde{\eta}_j, \qquad \hat{b} = \mathbf{1}[D>0], \tag{5}$$

so the payload signal grows as $RA$ while the random chips make the host terms average toward zero ($\mathrm{std} \propto \sqrt{R}$): the detection signal-to-noise ratio improves as $\sqrt{R}$. The per-bit magnitude $|D|$ is the soft confidence passed to the error-correction decoder. This is a textbook blind, spread-spectrum watermark detector, and it is what keeps the measured raw BER in §VII near $2\%$ through aggressive JPEG—low enough for the error-correcting code to recover the payload.

### F. Commitment and notarization

Immediately after readout, the sensor forms and signs a *capture record* (§IV-G) that commits, among other fields, to the raw frame $\mathcal{H}(I)$, to the camera's own encoded file $\mathcal{H}(E(I))$, and to the payload $\mathbf{m}$. The hard binding is the record hash

$$h = \text{SHA-256}\big(\, r \,\|\, \psi \,\big), \tag{6}$$

where $r$ is the record and $\psi$ the sensor's signature on it. For scale, many record hashes are aggregated into a Merkle tree [43] and only the root is written on-chain, amortizing gas cost; the contract records the block timestamp, which serves as the public "proof of existence" [11]. If connectivity is briefly unavailable, $h$ is queued in secure storage and submitted on reconnect, with the latency recorded.

### G. Device identity and content-bound records

The OTP identifier inside $\mathbf{m}$ names a sensor, but it proves nothing: it is a public number, and the in-pixel mark that carries it can be estimated from a single shared image and pasted onto other content without the carrier key (the classic copy attack [24]; §VII-C measures a $78\%$ success rate at the watermark level). Identity must therefore rest on a secret that only the sensor holds and that the ledger can vouch for. PIXELROOT adds four elements, all anchored on the same blockchain as the captures.

**Algorithm 1** PIXELROOT capture-time embed, sign, and notarize

**Require:** carrier key $k$, amplitude $A$, repetition $R$, basis $\Phi_{u,v}$; on-die key $sk$ with certified $pk \in$ lot $\Lambda$

$\beta \leftarrow$ (number, hash) of a recent block {capture-time lower bound}
ID $\leftarrow \mathrm{readOTP}()$ {48-bit pseudonymous handle}
$t \leftarrow \mathrm{attestedClock}()$; $(\phi, \lambda_{\mathrm{geo}}) \leftarrow \mathrm{gps}()$
$\mathbf{m} \leftarrow \mathrm{ID} \,\|\, t \,\|\, \mathrm{quant}(\phi, \lambda_{\mathrm{geo}})$ {128 bits}
$\mathbf{b} \leftarrow \text{ECC-Encode}(\mathbf{m})$ {RS, coded bits $b_1 \ldots b_{n_c}$}
$\mathrm{prng} \leftarrow \mathrm{CSPRNG}(\text{SHA-256}(k))$
**for all** coded bit $b_i$, $\sigma_i \leftarrow 2b_i - 1$ **do**
  draw $R$ distinct blocks $\{B_j\}$, chips $c_j \in \{\pm 1\}$ via prng
  **for all** $(B_j, c_j)$ **do**
    set per-region gain ramp on $B_j$: add $A\,\sigma_i c_j\,\Phi_{u,v}$
  **end for**
**end for**
$I \leftarrow \mathrm{expose_and_readout}()$; $E \leftarrow \mathrm{encode}(I)$
$r \leftarrow (pk, \mu, \Lambda, \mathbf{m}, \mathcal{H}(I), \mathcal{H}(E), F(I), \beta, c)$; $\psi \leftarrow \mathrm{Sign}_{sk}(r)$
$h \leftarrow \text{SHA-256}(r \,\|\, \psi)$; enqueue $h$ for Merkle batch
**on flush:** $root \leftarrow \mathrm{Merkle}(\{h\})$; $\mathrm{register}(root)$
**store** $E$ with $(r, \psi, \pi)$ in a C2PA manifest; publish $(r, \psi, \pi)$ to the record store when $E$ is published

*a) An on-die sensor key:* At manufacture each sensor generates an ECDSA P-256 key pair [34] inside the die—in a secure element on the stacked logic tier (M3, §IV-B) or derived from a physically unclonable function—so the private key $sk$ never leaves the sensor, not even to its manufacturer. The OTP identifier becomes a pseudonymous handle; the public key $pk$ is the credential. Because every key is unique to one die, physically extracting a key compromises one device, not a fleet.

*b) Certification and revocation on the ledger:* Manufacturers are admitted to the registry by its governance (in a standard, a consortium multisignature). A manufacturer $\mu$ certifies a production lot by registering the Merkle root $\Lambda$ of $\{\text{SHA-256}(pk_i)\}$; each sensor stores its inclusion path, so a lot of $10^5$ sensors costs one transaction. Lots are keyed by the pair $(\mu, \Lambda)$, so no other party can claim, or revoke within, a manufacturer's lot, even by registering the same root first. A manufacturer can revoke one of its sensor keys (a stolen or physically attacked device), and governance can revoke a manufacturer. Both are *time-scoped*: the ledger records the revocation time, captures sealed before it remain valid, and later ones fail. Revocation is enforced by the contract's own verification call, not left to validator policy—the gap that let revoked certificates keep validating after the Nikon incident [13].

*c) A signed, content-bound capture record:* After readout the sensor forms

$$r = \big(pk,\, \mu,\, \Lambda,\, \mathbf{m},\, \mathcal{H}(I),\, \mathcal{H}(E(I)),\, F(I),\, \beta,\, c\big) \tag{7}$$

and signs it, $\psi = \mathrm{Sign}_{sk}(r)$, where $c$ is a monotonic capture counter kept in the sensor. Committing $\mathcal{H}(E(I))$, the hash

of the camera's own encoded file, makes the hard binding checkable on the file people actually receive; $\mathcal{H}(I)$ keeps raw-domain forensics possible. $F(I)$ is a robust content fingerprint of the frame's *interior*: a border of $M = 32$ px—at least the largest translational offset the decoder searches—is excluded, the interior is divided into $8 \times 8$ tiles, each tile's luma is box-downsampled to $16 \times 16$ cells, and the $8 \times 8$ lowest DCT coefficients of each tile, taken relative to the interior's mean (so the DC term encodes the tile's brightness), are kept (8,448 bytes at int16). The fingerprint distance is the largest per-tile relative error,

$$d(F, F') = \max_{\ell} \frac{\|f_\ell - f'_\ell\|_2}{\|f_\ell\|_2 + \varphi}, \tag{8}$$

with an energy floor $\varphi$ so that flat tiles do not dominate. Because no tile is ever skipped and the excluded border is wider than any accepted shift, a crop cannot hide an edit inside the fingerprinted region. Compression, noise, mild filtering, rescaling and photometric edits leave every tile's coarse structure nearly unchanged; different content, or a foreign region large enough to change a tile, does not. The record ($\approx$18 KB as hex JSON) is published by the capturing party, the OEM, or a content-addressed store when the image is published; only its hash reaches the chain. The record store is untrusted: a lookup by payload may return several records (bursts captured within one second share $\mathbf{m}$), and every candidate is checked against the ledger.

*d) What a transcoded verdict means:* A fingerprint that survives JPEG $Q10$ must ignore detail below its resolution, and an adversary who holds the original can re-impose the original's cell means on an edited image (§VII-C). V2 therefore certifies *derivation*, not pixel integrity: TRANSCODED-CONSISTENT means that the questioned image carries a payload whose sensor-signed record describes the same scene at the fingerprint's resolution, and the verifier returns that record—and, when the capturing party publishes it, the original camera file—so that a viewer can compare the two. Only CAMERA-ORIGINAL (V1) certifies exact pixels. Copy attacks across scenes, large splices, and fabricated content are rejected; local edits within one capture are surfaced by comparison with the original, not excluded by V2.

*e) A capture-time bracket:* $\beta$ is the number and hash of a block the device observed before exposure. The contract records both the time and the block number at which each batch is sealed; a record whose anchor block is later than its sealing block, or whose anchor hash is not the canonical hash of that block, is rejected, and a valid anchor proves the record was formed after block $\beta$—a *lower* bound on capture time that complements the upper bound of Prop. 1 (Prop. 5).

*f) Why not ask the manufacturer to sign each capture:* An online, per-capture signature by the manufacturer would add no security over the sensor's own certified key, and it would cost a great deal: capture would fail offline (conflict zones, aircraft, courtrooms); the manufacturer would learn who captured what, when and where; and a breached, coerced, or defunct manufacturer would stop or forge its whole fleet's provenance. PIXELROOT therefore confines the manufacturer to certifying keys once and revoking them on the ledger; an optional manufacturer co-signature of batch roots can serve as a higher assurance tier, never as a requirement.

*g) The role of the carrier key:* With identity carried by the signed record, the carrier key $k$ protects the robustness of the channel, not its authenticity: a forger who knows $k$ can read or copy a payload but still cannot produce a record that a certified sensor signed for the forged content. An OEM may therefore publish its carrier key, which makes V2 verifiable by anyone—closing the gap between the publicly verifiable hard binding and the key-holder-verified soft binding (goal G4).

### H. Smart-contract registry

On-chain state is intentionally minimal and holds only ledger facts: admitted manufacturers, certified lots of sensor keys, revocation times, and the block time at which each batch root was sealed. Every Merkle tree (lots and capture batches) uses the same domain-separated construction, $\text{leaf} = \text{SHA-256}(\texttt{0x00} \| x)$ and $\text{node} = \text{SHA-256}(\texttt{0x01} \| \min(a,b) \| \max(a,b))$, with an odd node promoted rather than duplicated; the prefix stops an internal node from being presented as a leaf, and promotion rules out the duplicate-last-node mutation. Batching $N$ captures under one root makes per-image gas $O(1)$ amortized, and inclusion is proven with a $\lceil \log_2 N \rceil$-length path. Sealing is deliberately permissionless—any device, app, or batcher may submit, so no party can censor a capture—and the submitter carries no trust: authenticity comes from the sensor signature inside each record, checked against the certified lots. The full contract, an 18-case test suite, and the measured gas (§VII-C) are released. A capture is *notarized* once `register` mines. A verifier later calls `verifyCapture` (a view call, no gas) with the batch root, the record hash and its path, and the sensor's manufacturer, lot root, key hash and path; it learns whether the record is sealed, at what time and in which block, and whether that sensor was certified and unrevoked at that time.

### I. Verification

A verifier holds a questioned file (bytes $E'$, pixels $I'$), the public ledger, the record store, and the carrier key $k$. It needs no claimed payload and no manifest, and it runs Algorithm 2. Both checks end in the same record test: a record is *valid* if its signature verifies under its $pk$; its hash is included under a sealed root at block time $t_b$; the sensor was certified by $\mu$ in $\Lambda$ and unrevoked at $t_b$ (`verifyCapture`); and its anchor block precedes the sealing block and carries that block's canonical hash.

**(V1) Hard binding.** Look up $\mathcal{H}(E')$ among valid records' $\mathcal{H}(E(I))$ or $\mathcal{H}(I)$. A match shows that this exact file existed at $t_b$ (Prop. 1) and was committed in a record signed by a sensor certified at that time (Prop. 4). It is conclusive but brittle: any re-encoding breaks it.

**(V2) In-pixel binding.** Regenerate the carrier layout from $k$ and the frame's block count $N_B$; after a block-grid *resynchronization* search (§VI-A), project each carrier block onto $\Phi_{u,v}$ (Eq. 4), de-spread and combine (Eq. 5), and RS-decode $\hat{\mathbf{m}}$. Fetch the records whose payload is $\hat{\mathbf{m}}$, and accept if one of them is valid and the questioned content matches its

**Algorithm 2** PIXELROOT verification of a questioned file $(E', I')$

**Require:** ledger, record store, carrier key $k$, threshold $\tau$

**if** $\exists$ valid $r$ with $\mathcal{H}(E') \in \{\mathcal{H}(E(I)), \mathcal{H}(I)\}$ **return** CAMERA-ORIGINAL
$(\{B_j\}, \{c_j\}) \leftarrow \text{Select}(\text{SHA-256}(k), N_B, n_c, R)$
$\delta^\star \leftarrow \arg\max_\delta \sum_i |D_i(\delta)|$ {block-grid resync, §VI-A}

**for all** coded bit $i$ **do**
  $D_i \leftarrow \sum_j c_j \, \langle I'_{B_j+\delta^\star}, \Phi_{u,v} \rangle / \|\Phi_{u,v}\|^2$
  $\hat{b}_i \leftarrow \mathbf{1}[D_i > 0]$; $w_i \leftarrow |D_i|$ {soft conf.}
**end for**
$\hat{\mathbf{m}} \leftarrow \text{ECC-Decode}(\hat{\mathbf{b}}, \mathbf{w})$
**for all** $r \in \text{store.lookup}(\hat{\mathbf{m}})$ **do**
  **if** $r$ valid **and** $d(F(r), F(I'_{\delta^\star})) \le \tau$ **then**
    **return** TRANSCODED-CONSISTENT, $r$
  **end if**
**end for**
**return** UNVERIFIED {fail-safe}

fingerprint, $d(F(r), F(I'_{\delta^\star})) \le \tau$ (Eq. 8), computed on the image re-aligned by the resync offset $\delta^\star$. The fingerprint test is what makes the payload non-transferable: a payload copied onto other content resolves to a record describing different content.

Media is accepted at the strongest level the evidence supports: CAMERA-ORIGINAL (the exact file a certified sensor committed), TRANSCODED-CONSISTENT (derived from a notarized capture: it carries a payload whose sensor-signed record describes the same scene; the verifier returns the record so the original can be consulted), or the safe UNVERIFIED.

## V. SECURITY ANALYSIS

We first state the formal guarantees, then walk through the threat catalogue of §III.

### A. Formal guarantees

**Definition 1** (Forgery)**.** *A forgery is a file that the verifier (Alg. 2) accepts as* CAMERA-ORIGINAL *or* TRANSCODED-CONSISTENT*, yet whose content was not produced by a certified* PIXELROOT *sensor executing Alg. 1, up to the transformations the verifier is designed to tolerate.*

**Proposition 1** (Hard-binding soundness)**.** *Under a collision-resistant hash $\mathcal{H}$* (A4) *and an immutable ledger* (A3)*, passing V1 implies that the exact file $E^\star$ was committed in a record sealed at block time $t_b$, i.e. it existed at or before $t_b$. Producing a distinct $E^\star \neq E$ that passes V1 against a record sealed for $E$ requires a second preimage of $\mathcal{H}$ (generic work $\approx 2^{256}$ for SHA-256); an adversary who also controls the original capture needs only a collision (generic work $\approx 2^{128}$).*

*Proof sketch.* The leaf is $\mathcal{H}(\texttt{0x00} \| \mathcal{H}(r \| \psi))$ and $r$ contains $\mathcal{H}(E)$; `verifyCapture` recomputes the root from the leaf and its path and compares it with the sealed value. A different $E^\star$ with the same digest is a second preimage (or a collision); a different leaf with the same root is a collision of the domain-separated node hash; rewriting the sealed root violates A3. □

**Proposition 2** (Wrong-key read)**.** *Let the carrier layout be generated by a pseudorandom function keyed with $k$. To a party without $k$, the assignment of the $N_B$ blocks to the $n_c$ coded bits and their chip signs is computationally indistinguishable from uniform, so the bits it de-spreads from a marked image are independent of the payload; its probability of decoding the payload is then bounded by Prop. 3. Empirically, reading each of the* 24 *marked Kodak images with* 50 *wrong keys (*1200 *trials) gives mean BER* 0.4996 *and* $0/1200$ *correct recoveries.*

*Remark.* Prop. 2 bounds *reading* without the key; it does not bound *estimating* the mark. The detector statistic uses only the public basis $\Phi_{u,v}$, so an adversary can estimate the embedded signal from a single marked image and copy or subtract it without knowing $k$ (§VII-C). PIXELROOT's authenticity therefore rests on Prop. 4, not on the secrecy of the layout. (The released prototype seeds its layout generator with 32 bits; the proposition assumes the full-width keyed PRF a deployment uses.)

**Proposition 3** (Soft-binding false-accept)**.** *Model the read-back of* (4) *on content that was* not PIXELROOT*-marked as independent bit guesses with per-bit error $p \approx \frac{1}{2}$, so a byte (symbol) matches a given codeword by chance with probability $q = (1-p)^8 = 2^{-8}$. With a Reed–Solomon code of $n_{\mathrm{RS}}$ symbols correcting $t_{\mathrm{RS}}$ symbol errors, the decoder outputs a given payload with probability at most*

$$\epsilon \;=\; \sum_{j=0}^{t_{\mathrm{RS}}} \binom{n_{\mathrm{RS}}}{j} (1-q)^{\,j}\, q^{\,n_{\mathrm{RS}}-j}, \tag{9}$$

*the probability that at least $n_{\mathrm{RS}} - t_{\mathrm{RS}}$ bytes match by chance; it outputs* any *of $S$ payloads in the record store with probability at most $S\epsilon$. For RS$(48, 16)$ ($n_{\mathrm{RS}} = 48$, $t_{\mathrm{RS}} = 16$), $\epsilon \le 1.8 \times 10^{-65}$, so even $S = 10^{15}$ stored records leave $S\epsilon < 10^{-49}$.*

*Proof sketch.* The number of chance byte matches is $\mathrm{Binom}(n_{\mathrm{RS}}, q)$, and the decoder returns a payload only if at least $n_{\mathrm{RS}} - t_{\mathrm{RS}}$ match, giving (9); a union bound over the stored payloads gives $S\epsilon$. Real host content makes bits correlated rather than independent; the empirical counterpart is the $0/24$ false decodes on unmarked images (§VII-B). □

**Proposition 4** (Identity non-transferability)**.** *Assume an EUF-CMA-secure signature scheme* (A4)*, an immutable ledger* (A3)*, and on-die sensor keys* (A5)*. If the verifier returns* TRANSCODED-CONSISTENT *for a file $I^\star$, then a record $r$ exists that (i) was signed by a sensor certified on the ledger and unrevoked when $r$ was sealed, and (ii) commits a fingerprint with $d(F(r), F(I^\star)) \le \tau$. An adversary holding no unrevoked sensor key therefore obtains this verdict only by forging a signature (negligible probability) or by presenting content whose fingerprint is within $\tau$ of a genuinely captured frame—that is, a derivative of that capture at the fingerprint's resolution.*

*Proof sketch.* Alg. 2 returns TRANSCODED-CONSISTENT only for a valid record, whose signature verifies under a key whose hash `verifyCapture` places in a certified lot and finds unrevoked at the sealing time. Without that key, a valid $(r, \psi)$ for a new $r$ is an existential forgery. Reusing a genuine $(r, \psi)$ requires passing the fingerprint test against $F(r)$, which $r$ fixes and the signature protects. □

Prop. 4 makes an identity non-transferable *across scenes*; it does not make a capture immune to local edits. Content within $\tau$ of a genuine frame includes every benign transcoding, and it also includes edits that keep each fingerprint cell's mean, which an adversary holding the original can construct (§VII-C). This is why a transcoded verdict links to the original capture rather than certifying pixels (§IV-G, §VI-C).

**Proposition 5** (Capture-time bracket)**.** *If a record with anchor* $\beta = (b_\beta, \mathrm{hash}_\beta)$ *is accepted, was sealed in block* $b_s$*, and* $\mathrm{hash}_\beta$ *equals the canonical hash of block* $b_\beta$*, then the sensor signed the record after block* $b_\beta$ *was produced and no later than block* $b_s$*: its capture time lies in* $(\mathrm{time}(b_\beta), \mathrm{time}(b_s)]$.

*Proof sketch.* The upper bound is Prop. 1. For the lower bound, $\mathrm{hash}_\beta$ is unpredictable before block $b_\beta$ exists (A3, A4), so a record that contains it, signed by an on-die key (A5), cannot have been signed earlier. The contract stores each batch's sealing block number, and the verifier checks $\mathrm{hash}_\beta$ against the chain through any node (the EVM exposes only the last 256 block hashes on-chain). □

**Take-away.** Each accepted verdict rests on exponential barriers: the ledger and hash (Prop. 1) for the exact file; the sensor's signature, its on-ledger certification, and the content binding (Prop. 4) for derivatives; and the RS code (Prop. 3) against chance decodes. We now map each catalogued attack to these guarantees.

### B. Threat catalogue

Table II maps each attack in the threat model to the assumption it relies on and the guarantee that defeats it; the extended version discusses each in turn. Two entries deserve emphasis. A genuine sensor filming a screen produces a valid new record for the re-capture event: PIXELROOT attests that a certified sensor recorded this light at this time, not that the scene was real, and capture-time liveness signals (depth, focus distance) are future work. A stolen sensor key can sign arbitrary content until it is revoked; the damage is confined to one identity, and the capture counter helps detect cloning.

TABLE II
THREAT CATALOGUE MAPPED TO DEFENSES AND GUARANTEES.

| Attack | Why it fails | Assump. | Guar. |
|---|---|---|---|
| Deepfake / synth. | no valid record; fingerprint mismatch | A3–A5 | P1,P4 |
| Mark copy / transplant | record's fingerprint ≠ new content | A3–A5 | P4 |
| Metadata forge/strip | V2 reads $\mathbf{m}$; record resolves | A3,A4 | P3,P4 |
| Re-capture (screen) | new record; no backdating | A3–A5 | P1,P5 |
| Replay / pre-stage | bracket $(\mathrm{time}(b_\beta), \mathrm{time}(b_s)]$ | A3–A5 | P1,P5 |
| PRNU copy [17] | identity is a signature, not PRNU | A4,A5 | P4 |
| Large splice | tile fingerprint mismatch | A4 | P4 |
| Local edit (adaptive) | accepted only as derivative; V1 fails | A3,A4 | P1,P4 |
| Sensor-key theft | one identity; revocation stops new records | A3 | P4 |
| Rogue manufacturer | governance revocation; no retro-lots | A3 | P4 |
| Launder / regenerate | fails *safe* to UNVERIFIED | A4 | P3 |

P1–P5 = Props. 1, 2, 3, 4, 5.

## VI. ROBUSTNESS UNDER COMPRESSION

The central engineering tension is that the hard binding (V1) is *fragile* by design—ideal for a pristine original—while real-world distribution demands *semi-fragile* survivability (V2). We therefore analyze the in-pixel channel under lossy coding.

### A. Geometric edits and synchronization

Block-grid embedding is sensitive to operations that move the grid: cropping, translation, padding, and rescaling. PIXELROOT addresses this with an explicit *resynchronization* stage and is honest about its envelope. (i) **Rescaling**: because the carrier is a low-frequency coefficient, moderate down/up-sampling largely preserves it, though not losslessly: under a bilinear resize round-trip the raw BER rises from $0.016$ to $0.026$ at $0.75\times$ and $0.042$ at $0.5\times$, and RS decoding recovers the payload on $22/24$ and $16/24$ images respectively ($22/24$ for a combined JPEG $Q70 + 0.75\times$ attack; §VII). (ii) **Translation/crop**: the verifier searches a range of block-grid offsets $\delta$ and keeps the most confident decode (line 3 of Alg. 2). The search must respect the carrier's shape: $\Phi_{0,2}$ is constant down each column, so the confidence peak is several pixels wide vertically but only one pixel wide horizontally. PIXELROOT therefore searches every horizontal offset on a vertical stride of $4$, then refines vertically around the best ($15 \times 57 + 7 = 862$ decodes for $\pm 28$ px). (An earlier search on a stride of $4$ in both axes could step over the horizontal peak; on $2\%$ crops it decoded only $20/24$ images at raw BER $0.11$.) With the separable search, top-left border crops of $1\%$, $2\%$ and $3\%$ (shifts of up to 23 px) are recovered on $23/24$ images each at raw BER $\leq 0.020$; larger crops were not tested. These crops, like the resize attacks, return an image of the *original dimensions* (content shifted and the border padded, or scaled and scaled back). A crop or rescale that changes the frame's dimensions changes the number of blocks $N_B$ and therefore the carrier layout, and is not recovered by the offset search; supporting it requires a synchronization template that restores the original grid and size. A known synchronization template (a fixed low-frequency pattern) can extend this to scale/anchor estimation, as in the watermarking-sync literature. (iii) **Limits**: large arbitrary crops, rotation, and projective warps that are not covered by the offset/scale search are *not* recovered; critically, PIXELROOT then *fails safe*—the de-spread correlation collapses to noise and the verifier returns

UNVERIFIED rather than a false accept (we observe BER $\approx 0.5$ when sync fails). Heavy geometric editing therefore degrades to "cannot verify," never to "falsely verified," and the hard binding (V1) plus perceptual-hash "same-scene" layers remain available.

### B. Video: MPEG / H.264 / HEVC

Video adds motion-compensated prediction, integer transforms, in-loop deblocking, and GOP structure [44], [45]. The video watermarking literature shows that embedding survives best in mid-frequency transform coefficients and that I-frame perturbations propagate ("drift") through a GOP, so embedding is often confined to portions of the GOP [46]–[48]. For PIXELROOT, the practical design is: embed the per-frame (or per-keyframe) payload at capture, and commit a Merkle root over keyframe hashes so that *frame-level* verification localizes tampering. Learned, codec-aware embedding has recently been shown to survive non-differentiable H.264 [48] and is a natural upgrade path.

### C. The semantic-authenticity boundary

A subtle limitation deserves emphasis. A benign platform filter may alter carrier pixels enough to fail V2 even though the image's *meaning* is unchanged—a false alarm—while a malicious edit might, in principle, preserve carriers. PIXELROOT therefore certifies *consistency with a notarized original*—exactly for V1, and up to the fingerprint tolerance $\tau$ for V2—not semantic equivalence. A local edit that keeps every tile within $\tau$ of the original (for example, a small object removed from a textured region) passes V2; edits that change a tile's coarse content, such as the 6–12% splices of §VII-C, do not. We argue this is the right primitive: it cleanly separates "bit-for-bit original" (V1), "faithfully transcoded" (V2), and "cannot establish provenance," and it composes with perceptual-hash and content-similarity layers [31] for softer "same-scene" judgments.

## VII. EVALUATION AND REFERENCE IMPLEMENTATION

### A. Reference implementation

We provide an open software prototype of the PIXELROOT core (payload layout, SHA-256 seeding, PRNG carrier selection, gain-value generation, image hashing, and a register/verify service with simulated notarization). It implements both a legacy v1 (120-bit) and the current v2 (128-bit) payload, the latter fixing a coordinate-packing precision bug. The prototype validates the encode/verify round-trip and the metadata→seed→position determinism end-to-end before silicon. A second, *real-image* harness (Python; `numpy`, Pillow's libjpeg/WebP, `scikit-image`, and a real `reedsolo` code) runs the full evaluation below on the public Kodak set; a claims-to-experiments matrix and reproduction instructions are released in `experiments/EXPERIMENT_PLAN.md` at https://github.com/mihirkavi/PixelRoot. The device-identity layer is implemented as a Solidity registry (`contracts/`, 18 Hardhat tests) and a Python reference of the sensor key, signed records, content fingerprint, and verifier (`ledger.py`, `device_identity.py`, 21 unit tests), with Merkle test vectors shared between the two implementations.

### B. Measured robustness of the in-pixel channel

We evaluate the spread-spectrum embedding (§IV-E) on *real public images* with *real image codecs*, following the current watermark-robustness benchmarking practice of WAVES [49] (a broad attack suite reported as performance-vs-quality with PSNR/SSIM). **Methodology.** The 24 Kodak natural photographs ($768 \times 512$) [50] carry a 128-bit payload protected by a real Reed–Solomon code $\mathrm{RS}(48, 16)$ (`reedsolo`); the 384 coded bits are spread with repetition over $8 \times 8$ blocks at amplitude $A = 4$ gray levels on the $(0, 2)$ carrier, embedded in the luma channel. Attacks use *actual libjpeg / WebP* (with 4:2:0 chroma subsampling), Gaussian noise and blur, bilinear resize round-trips, photometric (brightness/contrast/gamma) edits, translational crop with grid resynchronization, and rotation/ resized-crop. Decoding uses the host-suppressed correlation de-spread of Eq. (5) followed by RS decoding. The harness is released (§VII, `experiments/`).

**Fidelity.** Embedding is imperceptible: mean PSNR 38.2 dB and SSIM 0.942 between original and marked frames (the per-frame fidelity rises to $\sim 41$ dB at $A = 3$). **Robustness.** Table III reports the raw (pre-ECC) coded-bit BER and the *end-to-end* payload-decode success after RS correction. On real content the raw channel is not error-free—mean BER $\approx 1.6\%$, content-dependent up to 13% on the most textured frame—but the RS code recovers the exact 128-bit payload on 22–23/24 images through real JPEG from $Q = 95$ down to $Q = 20$ (and 21/24 at the extreme $Q = 10$), Gaussian noise to $\sigma = 20$, light blur ($\sigma = 0.5$), and translational crops of 1–3% with resync, and on 21–22/24 under WebP, resize to 0.75–1.5×, photometric edits, and combined JPEG+resize. Recovery is weaker under stronger blur ($\sigma = 1.0$: 19/24) and 0.5× resize (16/24). Failures are honest and informative: a single very-high-frequency frame exceeds the ECC capacity at $A \leq 6$, and *geometric* rotation ($\pm 1$–$2^\circ$) and resized-crop drive BER to $\approx 0.5$—a documented limitation requiring explicit rotation/scale synchronization (§VIII). **Operating curve.** An amplitude sweep at a fixed strong attack (JPEG $Q40$) over all 24 images, exposing a genuine cliff: post-ECC decode rises from 0/24 at $A = 0.5$ to 23/24 at $A = 4$ (38 dB), so $A = 4$ is the imperceptible-yet-robust operating point. **Security / false-accept.** Reading a marked image with 50 wrong keys across all images (1200 trials) gives mean BER 0.4996 and 0/1200 recoveries; running the correct-key decoder against the 24 *unmarked* images (a deepfake stand-in) against a claimed payload gives mean BER 0.494 and 0/24 false accepts. These match the $p \approx \frac{1}{2}$ model of Props. 2–3, whose closed-form RS false-accept probability is $\leq 1.8 \times 10^{-65}$ for this code. A *copy* attack, however, succeeds at the watermark level: lifting the marked residual from image $A$ onto a different image $B$ makes $B$ decode to $A$'s payload in 15/17 cases. The in-pixel channel alone does not make an identity non-transferable; §VII-C shows that the signed, content-bound record does.

### C. Device identity and content binding (E6)

We evaluate the full verification path of §IV-G on the same 24 Kodak images. **Setup.** One manufacturer admitted to the

TABLE III
MEASURED IN-PIXEL-CHANNEL ROBUSTNESS ON **24 REAL KODAK IMAGES** WITH **REAL LIBJPEG/WEBP** CODECS (128-B PAYLOAD, $\mathrm{RS}(48,16)$, $A=4$; PSNR 38.2 DB, SSIM 0.942). "RAW BER" IS THE PRE-ECC CODED-BIT ERROR; "DECODE" IS EXACT POST-ECC PAYLOAD RECOVERY.

| Attack | raw BER | decode |
|---|---|---|
| JPEG $Q95$–$Q20$ | 0.016 | 22–23/24 |
| JPEG $Q10$ | 0.022 | 21/24 |
| WebP $q80$, $q60$ | 0.021–0.024 | 21–22/24 |
| Gaussian noise $\sigma \leq 20$ | 0.017 | 23/24 |
| Gaussian blur $\sigma = 0.5$ | 0.018 | 23/24 |
| Gaussian blur $\sigma = 1.0$ | 0.033 | 19/24 |
| Resize $0.75\times$, $1.5\times$ | 0.025 | 22/24 |
| Resize $0.5\times$ | 0.042 | 16/24 |
| Bright/contrast/gamma $\pm 10\%$ | 0.017 | 21–23/24 |
| Crop 1–3% + resync | 0.019 | 23/24 |
| JPEG$Q70$ + resize $0.75\times$ | 0.026 | 22/24 |
| JPEG $Q5$ (beyond range) | 0.221 | 0/24 |
| Rotation $\pm 1$–$2^\circ$ *[geom. limit]* | 0.50 | 0/24 |
| Resized-crop *[geom. limit]* | 0.49 | 0/24 |
| *Wrong-key read* (1200 trials) | 0.4996 | 0/1200 |
| *Forgery / unmarked* (24 trials) | 0.494 | 0/24 |
| *Copy attack* $\rightarrow$ source payload | — | 15/17 |

TABLE IV
E6 GENUINE-MEDIA VERDICTS ON THE **12 HELD-OUT KODAK IMAGES** (CO = CAMERA-ORIGINAL, TC = TRANSCODED-CONSISTENT, U = UNVERIFIED); $\tau$ CALIBRATED ON THE OTHER 12. EVERY U IS A PAYLOAD-DECODE FAILURE.

| Condition | CO | TC | U |
|---|---|---|---|
| Camera JPEG (exact file) | 12 | 0 | 0 |
| JPEG $Q95$–$Q20$ | 0 | 11 | 1 |
| JPEG $Q10$; WebP $q60$ | 0 | 10 | 2 |
| WebP $q80$; noise $\sigma \leq 20$; blur $\sigma \leq 1$ | 0 | 11 | 1 |
| Resize $0.75\times$, $1.5\times$ | 0 | 11 | 1 |
| Resize $0.5\times$ | 0 | 9 | 3 |
| Bright./contrast/gamma $\pm 10\%$ | 0 | 10–11 | 1–2 |
| Crop 1–3% + resync | 0 | 11 | 1 |
| JPEG+resize; JPEG+noise | 0 | 11 | 1 |
| All processed copies | 0 | 358 | 38 |
| Unmarked originals | 0 | 0 | 12 |

registry certifies a lot of eight sensors, each holding an on-die ECDSA P-256 key; every image is marked as in §VII-B ($A=4$, $\mathrm{RS}(48,16)$), encoded by the "camera" as JPEG $Q95$, and described by a sensor-signed record (Eq. 7); the record hashes are sealed as one Merkle batch. The ledger is an executable model of the contract of §IV-H; the contract itself passes an 18-case test suite on the Hardhat EVM covering the same state machine, Merkle compatibility with the Python implementation through shared test vectors, the leaf/node domain separation, the per-manufacturer lot keys (including a front-running race), the sealing block number, and every revocation rule. To avoid tuning on the test set, the fingerprint threshold is calibrated on the 12 even-indexed images (midpoint between the largest genuine distance under the benign suite of Table III and the smallest distance of 25–35% splices), and *every result below is on the* 12 *held-out images*, with Wilson 95% intervals where rates are reported.

**Fingerprint margin.** Calibration gives $\tau = 0.482$. On held-out images the largest genuine distance under any benign attack is 0.197, while different images are at least 1.085 apart (median 2.55).

**Genuine media** (Table IV). The camera's own JPEG verifies as CAMERA-ORIGINAL on $12/12$ images through V1 alone, with no dependence on the watermark. Processed copies verify as TRANSCODED-CONSISTENT in 358 of 396 cases (90%, CI 0.87–0.93): $11/12$ images across JPEG down to $Q20$, noise, blur, resize 0.75–1.5×, photometric edits, 1–3% crops and combined attacks, and 9–$10/12$ under JPEG $Q10$, WebP $q60$, brightness $\times 1.1$ and $0.5\times$ resize. All 38 UNVERIFIED genuine verdicts are payload-decode failures (mostly the texture-heavy frame of §VII-B); the fingerprint rejected none. None of the 12 unmarked originals is accepted.

**Cross-scene forgeries** (Table V). The keyless copy attack estimates the mark from a JPEG $Q75$ copy of the source by the detector's own public projection and high-pass and adds $1.5\times$ the estimate to another image; the forgery decodes the source's payload in $61/78$ attempts (78%), and $70/78$ with the original in hand. The content-bound record rejects all 156 (CI upper bound 0.05 per attack), and all 24 splices covering 6–12% of the frame, including the 21 whose payload still decodes.

**Local edits within one capture.** Smaller pasted patches are sometimes within $\tau$: $3/12$ at $64 \times 64$ px (about 1% of the frame), $1/12$ at $96 \times 96$, $0/12$ at $128 \times 128$, and $2/12$ for a 128-px patch near the border followed by an 8-px shift (a whole foreign tile row followed by the same shift is rejected $12/12$, so a crop no longer hides content). An *adaptive* adversary who holds the original does better: after pasting a $96 \times 96$ foreign patch it re-imposes the original's mean on every fingerprint cell, and $11/12$ such images are accepted. The pasted detail survives only below the cell size (about 5 px here): its correlation with the donor is 0.52, and it changes the patch by 17.8 gray levels on average—visible on inspection (Fig. 2). These results are why PIXELROOT defines a transcoded verdict as *derivation from a notarized capture*, returns the record (and, when published, the original file) for comparison, and reserves pixel-exact certification for V1 (§IV-G).

**Ledger and record attacks.** All eleven record- and ledger-level cases in Table V behave as specified, including time-scoped revocation in both directions, an anchor block later than the seal, an anchor hash that is not the chain's, and a burst of two captures sharing one payload (both resolve).

**Cost.** Signing a capture (fingerprint and ECDSA) takes 9.0 ms in Python on a commodity CPU; ECDSA verification takes 0.15 ms, and verifying an exact camera file end to end 4.2 ms. A record is $\approx$ 18 KB as hex-encoded JSON (about half that in binary), off-chain. On-chain (Hardhat EVM, `solc` 0.8.24), sealing a batch costs 47,016 gas regardless of its size—4.70 gas per image at $N = 10^4$—certifying a lot costs 51,626 gas, revoking a sensor 65,859 gas, and verification is a free view call.

**Scope and threats to validity.** These results validate the *algorithmic* core—carrier design, host-

TABLE V
E6 ATTACKS ON THE 12 HELD-OUT IMAGES: WATERMARK-LEVEL SUCCESS VERSUS ACCEPTANCE AS TRANSCODED-CONSISTENT BY ALG. 2.

| Attack | Payload decodes | Accepted |
|---|---|---|
| *Cross-scene* | | |
| Keyless copy (from JPEG $Q75$) | 61/78 | 0/78 |
| Copy with original residual | 70/78 | 0/78 |
| Splice 6–12% of frame, $Q90$ | 21/24 | 0/24 |
| *Local edits within one capture* | | |
| Patch 64 px, $Q90$ | 11/12 | 3/12 |
| Patch 96 px, $Q90$ | 11/12 | 1/12 |
| Patch 128 px, $Q90$ | 11/12 | 0/12 |
| Foreign tile row + 8-px shift | 8/12 | 0/12 |
| 128-px patch near border + shift | 11/12 | 2/12 |
| Adaptive: 96-px patch + cell means | 11/12 | 11/12 |
| *Record and ledger* | | |
| Record edited (fingerprint, file hash) | — | 0/2 |
| Uncertified sensor key; borrowed signature | — | 0/2 |
| Revoked sensor, sealed after revocation | — | 0/1 |
| Revoked sensor, sealed before revocation | — | 1/1 (valid) |
| Anchor after seal; forged anchor hash | — | 0/2 |
| Record not notarized | — | 0/1 |
| Burst: two records, one payload | — | 2/2 (valid) |



Fig. 2. Adaptive local edit accepted as TRANSCODED-CONSISTENT (E6). Left: marked original. Centre: a $96\times96$ patch from another photograph, pasted and then forced to the original's fingerprint-cell means, after JPEG $Q90$. Right: the donor. The edit is visible on inspection and appears in a comparison with the original that the record links to; only V1 certifies exact pixels.

suppressed spread-spectrum detection, robustness to *real* JPEG/WebP/noise/blur/resize/photometric/crop attacks, false-accept behavior, and tamper localization—on *real public imagery* with *real codecs* and a *real Reed–Solomon* code, independent of silicon. They embed an *additive* luma perturbation, whereas a sensor gain change is *multiplicative*: the imprinted signal scales with scene brightness, weakening in dark regions and clipping at saturation. They do not yet model this, demosaicing from the Bayer/RAW domain, real per-column gain calibration error, lens/optical effects, or the analog photo-electron domain. The device-identity results (E6) use a software model of the sensor key and an executable model of the ledger; the contract is tested on a local EVM, not on a public network; the PRNU comparison uses real Kodak *content* with the textbook multiplicative sensor model (per-camera RAW datasets are access-gated). The full harness, datasets script, and figures are released for reproduction.

### D. Remaining silicon- and codec-dependent validation

The following require hardware or a full pipeline and are future work: (F1) RAW/Bayer-domain capture and a multi-camera PRNU study on Dresden/VISION; (F2) video robustness vs. H.264/HEVC QP and GOP placement with keyframe-level Merkle commitments; (F3) an end-to-end re-capture and PRNU-copy [17] study, including capture-time liveness fields; (F4) shutter-to-inclusion latency and finality on a public network (gas is measured, §VII-C); (F5) a silicon study of per-region/per-pixel gain modulation on a sensor evaluation board, including multiplicative gain on RAW data through a full ISP; and (F6) on-die key generation and signing in a sensor secure element.

## VIII. DISCUSSION: LIMITATIONS, PRIVACY, STANDARDIZATION

**Hardware dependency and assurance tiers.** PIXELROOT's strongest guarantees require per-column/per-region gain and secure-element support (§IV-A). We make the trust gradient explicit rather than hiding it, reporting one of three tiers with every verdict: *Tier A (sensor-rooted)*—modulation imposed in the photo-electron domain by gain hardware; *Tier B (ISP-attested)*—the identical modulation applied in an attested ISP on legacy devices, trustworthy only up to the attestation; and *Tier C (software/notarize-only)*—a C2PA/ProofMode-style capture-time hash with no in-pixel binding. Verifiers and policies must not conflate tiers.

**Verification trust model.** Every check in Alg. 2 uses public information. The hard binding (V1) needs only the file, the record and the ledger. The in-pixel binding (V2) additionally needs the carrier key $k$ to locate the carriers; because authenticity rests on the sensor's signature and the content binding (Prop. 4) rather than on the secrecy of $k$, an OEM can publish its carrier key and let any platform, court, or viewer run V2. An OEM that prefers to keep $k$ private (to make casual removal of the mark slightly harder) can instead operate a verification service; the security of an accepted verdict is the same either way.

**Privacy-preserving notarization.** No personal data reaches the chain: it holds only lot roots of sensor-key hashes, revocations, and opaque Merkle roots of record hashes. Records live off-chain, are published only for images their owner publishes, and can be erased; location is optional and off by default; the in-pixel identifier is a rotating pseudonymous handle. A published record does reveal its sensor key, so two published images from one sensor are linkable; anonymous-attestation signatures [51] are the natural upgrade.

**Why a public ledger rather than a timestamping authority?** An RFC 3161 authority or a transparency log [52] gives append-only timestamps, but each is a party that can be coerced, compromised, or censored—and certificate compromise has already forced wholesale revocation in a deployed C2PA camera [13]. A public chain offers permissionless auditability and censorship resistance, and here it also holds sensor certification and revocation, so "was this sensor trusted when this capture was sealed?" has one public answer. Merkle batching makes the cost $O(1)$ per capture (47,016 gas per

batch); an authority or log can serve as a fast path anchored on the chain.

**Failure-safe semantics.** As emphasized, PIXELROOT declines rather than falsely certifies under heavy laundering or unsynchronizable geometric edits; downstream policy must treat "unverified" as "unknown," not "fake."

**Adoption and standardization.** PIXELROOT cannot verify media from cameras that never marked it; unmarked media is labeled "unverified," never "fake." We propose a phased rollout (software capture apps, then attested ISPs, then sensor gain hardware) and a cross-OEM profile that fixes the payload, carrier, record format, fingerprint, and registry interface, reusing C2PA manifests as transport.

## IX. Conclusion and Future Work

Reactive deepfake detection cannot win a generator-vs-detector arms race, and metadata-only provenance breaks the moment a credential is stripped. PIXELROOT relocates the root of trust into the pixels and onto a public ledger: an error-corrected payload embedded by the CMOS sensor at capture, a capture record signed by a key that only that sensor holds, and a blockchain that certifies the sensor, seals the record, and records any revocation. We showed that the in-pixel mark alone is not enough—it can be copied onto other content without any key—and that binding it to a sensor-signed content fingerprint makes the identity non-transferable: on held-out images no cross-scene copy or large splice was accepted, while $90\%$ of processed genuine copies verified through compression, resizing, photometric edits and crops. We also showed the limit of any tolerant binding: an adversary who holds the original can make local edits that keep the fingerprint, so a transcoded verdict means *derived from* a notarized capture, which the verifier returns for comparison, and only the camera-original verdict certifies exact pixels. We are careful about what this paper establishes: the algorithmic core and the identity layer are *implemented and measured* on real public images with real codecs; the registry contract is *implemented and tested* on a local EVM with measured gas, but not yet deployed on a public network; and the sensor-level gain hardware and on-die keys are a *feasibility analysis* grounded in shipping process technology, not a silicon result. Future work includes silicon-level validation on sensors with per-region gain control, codec-aware learned embedding for video with per-segment records and Merkle trees of segments [48], capture-time liveness fields against screen re-capture, anonymous-attestation signatures for unlinkable records, and pursuing PIXELROOT as an open cross-OEM standard. We view this as a foundational—if not all-encompassing—step toward driving deepfake-driven disinformation toward practical irrelevance for media that matters.

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## Conflict of Interest

The author declares that there is no conflict of interest regarding the publication of this article. The work received no external funding, and the author has no financial or personal relationships that could have inappropriately influenced it.

## Reproducibility and Disclosure

The reference implementation and the evaluation harness (https://github.com/mihirkavi/PixelRoot)—the keyed spread-spectrum embed/extract of §IV-E, the Reed–Solomon error-correction layer, the WAVES-aligned attack suite, and the tamper-localization and PRNU pipelines, the device-identity reference and verifier, and the registry contract with its test suite—are part of the open PIXELROOT project. All measured numbers in §VII were produced on the public Kodak image set with production codecs (`libjpeg` JPEG and `libwebp` WebP) and are reproducible end-to-end from the released scripts, a `requirements.txt`, and the contract's locked `npm` dependencies (the Solidity compiler is pinned to 0.8.24). The ledger in the Python experiments is an executable model of the contract, and the contract is tested on a local EVM; neither has been deployed to a public network. The sensor keys are software stand-ins for on-die keys, and the silicon-dependent per-pixel gain mechanisms of §IV-B are a feasibility analysis grounded in the cited fabrication literature, not fabricated results.

## Data Availability

The Kodak TrueColor image suite used in the experiments is publicly available. The provenance payloads are synthetic and contain no personal data. No human or animal subjects were involved in this work.

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